

Chien-Wen (Wayne) Pan

Email: cwpan@umn.edu | Phone: (+1) 734-780-0168

SUMMARY

Postdoctoral Associate working on an upper-limb robot from zero to demo for a university spinout company. During the doctoral study, my work utilizes Koopman-inspired methods and physics-informed machine learning, bridging biomechanical and physiological signals with human intentions for wearable robots and healthcare applications. Experienced in modeling and characterizing of electromechanical actuators, traditional and modern control algorithms and simulations for robots, and multi-modal sensory. Passionate about developing sensory intelligence and adaptive robotics systems.

SKILLS

Domain Robotics simulation, Reinforcement learning, Nonlinear modeling, System identification, Signal processing, Physics-informed machine learning, Mechatronics

Programming and Tools: Python, PyTorch, TensorFlow, MuJoCo, C/C++, MATLAB/Simulink, Git/Github, Linux, ROS2

Hardware and Prototyping: Embedded systems and edge compute (Raspberry Pi, Jetson nano, Arduino, STM32), 3D printing, laser cutting, Solidworks, LabVIEW

EXPERIENCE

University of Minnesota Twin Cities (Human Sensorimotor Control Lab)

Minneapolis, MN

Postdoctoral Associate

Mar. 2026 – Present

- Co-authored a \$1.25M NSF SBIR Phase 2 grant proposal based on successful hardware and software prototype validation.
- Integrated BLDC actuators and sensors with STM32 boards, implementing CAN bus communication between parent and child nodes for an upper-limb robot.
- Designed closed-loop control algorithms and integrated the hardware with Unity-based gamified UI.
- Conducted 100+ comprehensive customer discovery interviews with stroke survivors, clinical directors, and physical therapists for NSF I-Corps summer 2026 cohort.

Arizona State University (Robotics and Dynamical Systems Group)

Tempe, AZ

Research Assistant / Ph.D. Candidate

Aug. 2021 – Nov. 2025

- Developed a two-stage Koopman-based pipeline predicting lower-limb kinematics from EMG signals, achieving 95% accuracy and stable near-future multi-step prediction. (This work is published in IEEE TBME 2026.)
- Trained a PIML model for BP estimation using PPG signals, integrating cardio-dynamics with nonlinear lifting; validated on 120 subjects in ICU condition. (This work is published in Biomedical Signal Processing and Control 2026.)

University of Michigan (Neurobionics Lab)

Ann Arbor, MI

Graduate Independent Research Study

Sep. 2017 – Jun. 2018

- Designed and built a power test bench to characterize a high-performance, exterior-rotor BLDC motor and drive system for an open source bionic leg. (This work is published in IEEE IROS 2019.)

INDEPENDENT ONGOING RESEARCH

Independent Researcher

Model-Predictive Control in MyoSuite simulation

Jan. 2026 – Present

- Expanding doctoral EMG-based state estimation framework to incorporate model-predictive control within the MuJoCo simulation environment.

PUBLICATIONS (SELECTED)

- C.W. Pan, M. Rahmani, and S. Redkar, 2026, “EMG-based gait estimation and prediction using Koopman-inspired method”, *IEEE Transactions on Biomedical Engineering*.
- C.W. Pan, M. Rahmani, and S. Redkar, 2026, “Integrating biophysical dynamics with data-driven method for blood pressure (BP) estimation using photoplethysmography (PPG)”, *Biomedical Signal Processing and Control*.
- U.H. Lee, C.W. Pan, and E.J. Rouse, 2019, “Empirical characterization of a high-performance exterior-rotor type brushless DC motor and drive.” *Proc. of 2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*.

EDUCATION

Arizona State University

Tempe, AZ

Ph.D., Systems Engineering

Dec. 2025

Dissertation: Model-based Machine Learning for Physiological Wearables in Healthcare and Human-Robot Interactions Committees: Sangram Redkar, Thomas Sugar, Kevin Nicols

University of Michigan

Ann Arbor, MI

M.S., Robotics

Apr. 2018

National Cheng-Kung University

Taiwan

B.S., Aeronautics and Astronautics

Jun. 2014